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CHENNAI - 602105

Assignment Answer Booklet

Advanced Development Methodologies, Cross-Disciplinary Project
Management, and Ethics in Software Engineering – UNIT I

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Question Attempted : Problem 3 – Lean + XP in a Real-Time Tutoring Platform

Scenario Recap

A team developing a real-time tutoring platform wants to reduce waste using Lean principles and improve quality using Extreme Programming (XP). The team must identify sources of waste in the current process, select appropriate XP practices, design a combined Lean + XP workflow, and produce a representative user story with acceptance criteria.

Task

Identify wastes, choose XP practices, design a combined workflow, and write a user story with an acceptance test.

1. Identification of Lean Wastes in the Tutoring Platform Project

Lean software development adapts the seven manufacturing wastes identified in the Toyota Production System to knowledge work. Applying this lens to a real-time tutoring platform (live video sessions, whiteboard sharing, scheduling, and payments) surfaces the following categories of waste.

Lean Waste Type	Manifestation in the Tutoring Platform
Partially Done Work	Half-built video-chat modules sitting in a branch for weeks while the scheduling module is prioritized
Extra Features	Building an in-app whiteboard with advanced vector drawing tools before confirming tutors actually want it
Relearning	Developers repeatedly re-discovering how the WebRTC signalling layer works because there is no documentation
Handoffs	Requirements passed from product owner to business analyst to developer through static documents
Delays	Waiting for a third-party payment gateway sandbox approval before the booking flow can be tested
Task Switching	A single developer split across the tutoring platform and an unrelated internal tool, reducing focus and productivity
Defects	Session-drop bugs discovered only after release because there was no automated regression suite

Of these, **partially done work**, **handoffs**, and **defects** are the highest-impact wastes for a real-time platform, because unfinished streaming code cannot be validated until integrated, and undetected defects directly damage the live-session user experience.

2. Extreme Programming (XP) Practices Selected

To directly counter the wastes identified above, the team adopts a focused subset of XP practices rather than the full XP catalogue, so that adoption remains realistic for a small team.

- **Pair Programming** – two developers work together on the WebRTC and scheduling modules, reducing relearning and defects by sharing tacit knowledge in real time instead of through documents.
- **Test-Driven Development (TDD)** – unit and integration tests are written before the streaming and booking logic, directly targeting the defects waste by catching regressions before release.
- **Continuous Integration (CI)** – every commit is built and tested automatically, which prevents partially done work from silently diverging from the main branch for weeks.
- **Small Releases** – the whiteboard and chat features are released incrementally to a pilot group of tutors, avoiding the extra-features waste by validating real demand before further investment.
- **On-Site Customer / Product Owner Availability** – a tutoring domain expert is available to answer questions directly, cutting out the handoff chain between analyst and developer.
- **Collective Code Ownership** – any pair can modify any part of the codebase, reducing the impact of task switching and bus-factor risk.

3. Combined Lean + XP Workflow

The workflow below fuses Lean's pull-based flow and waste elimination with XP's engineering discipline. Work is pulled from a prioritized backlog only when the team has capacity (Lean), and every unit of work passes through pairing, TDD, and continuous integration (XP) before it is released in small batches and validated with fast customer feedback.

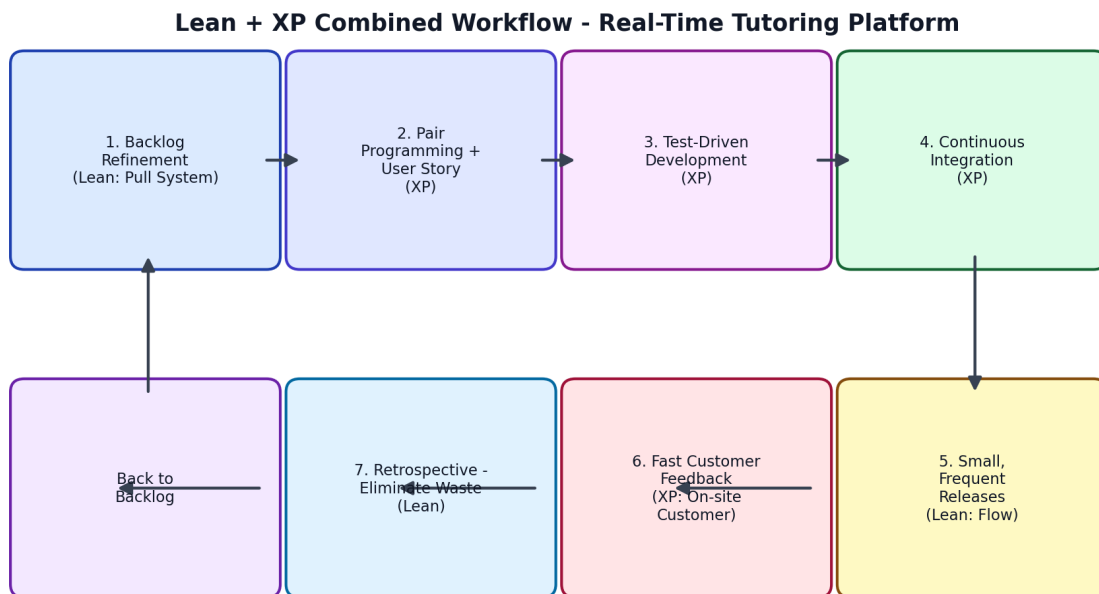


Figure 1: Combined Lean + XP workflow for the real-time tutoring platform, showing the cycle from backlog pull through release, feedback, and waste-focused retrospective.

Workflow Narrative

- **Backlog Refinement (Lean – Pull System):** Only the highest-priority, thinnest vertical slice of a feature (e.g., "tutor can share a whiteboard") is pulled into progress, keeping work-in-progress low.
- **Pair Programming + User Story (XP):** A pair takes the story, clarifies acceptance criteria with the product owner, and begins implementation together.
- **Test-Driven Development (XP):** Failing tests are written first for the session-join and annotation logic, then the minimum code needed to pass them is added.
- **Continuous Integration (XP):** Code is merged and built multiple times a day; automated tests run on every push to catch defects immediately.
- **Small, Frequent Releases (Lean – Flow):** The completed slice ships to a pilot cohort of tutors within days, not months, keeping inventory of unreleased work near zero.
- **Fast Customer Feedback (XP – On-site Customer):** Tutors and students give direct feedback on session quality, which feeds straight back into the backlog.
- **Retrospective – Eliminate Waste (Lean):** The team explicitly reviews which of the seven wastes appeared that iteration and agrees on one concrete improvement action.

4. User Story with Acceptance Criteria

User Story

As a tutor running a live session, *I want to* share an interactive whiteboard with my student in real time, so *that* I can explain concepts visually without leaving the video call.

Acceptance Criteria (Gherkin-Style)

- **Given** a tutor and student are both connected to an active session, **when** the tutor clicks "Open Whiteboard", **then** a shared whiteboard canvas appears for both participants within 2 seconds.
- **Given** the whiteboard is open, **when** the tutor draws or writes on the canvas, **then** the student's view updates in near real time with a latency under 300 milliseconds.
- **Given** a whiteboard session is in progress, **when** the network briefly drops and reconnects, **then** the whiteboard state is automatically resynchronized without data loss.
- **Given** the session ends, **when** the tutor closes the whiteboard, **then** a snapshot image is saved and made available for both users to download afterward.

This story is deliberately small enough to fit within one or two pairing sessions, matching the Lean principle of thin vertical slices and the XP principle of frequent, testable increments. Each acceptance criterion above maps directly to an automated test written under the TDD practice described in Section 3, closing the loop between Lean flow and XP engineering rigor.

5. Why Combining Lean and XP Works for This Platform

Lean and XP are complementary rather than overlapping: Lean operates at the level of the overall flow of work through the team, asking whether the right thing is being built at the right time without waste, while XP operates at the level of the engineering practices used to build each unit of work, asking whether it is being built correctly, safely, and verifiably. For a real-time tutoring platform, where session quality issues are visible to paying customers within seconds, this combination is particularly valuable.

The Lean pull system keeps work-in-progress low, which means the team is never trying to finish five half-built features at once — a common source of the partially done work waste identified in Section 1. XP's pairing and TDD practices then ensure that whatever is pulled into progress is built with a safety net, so the small, frequent releases that Lean encourages do not become a source of new defects. In effect, Lean decides what to build next and how much to build at once, while XP decides how to build it reliably.

The retrospective step closes the loop: instead of a generic open discussion, the team explicitly checks the seven Lean wastes against the completed iteration. For example, if the retrospective reveals repeated relearning about the WebRTC signalling layer, the very next iteration's improvement action might be to pair-write a short internal guide, directly converting a Lean observation into an XP-compatible corrective practice.

6. Summary

- Seven Lean wastes were mapped to concrete situations in the tutoring platform, with partially done work, handoffs, and defects flagged as highest priority.
 - Six targeted XP practices — pair programming, TDD, continuous integration, small releases, on-site customer, and collective ownership — were selected specifically to counter those wastes.
 - A seven-step combined Lean + XP workflow was designed and illustrated in Figure 1, cycling from backlog pull through release, feedback, and a waste-focused retrospective.
 - A representative user story for shared whiteboarding was written with four Gherkin-style acceptance criteria that map directly onto TDD test cases.
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